PulseWatch (pingbud.us)

Complete Platform Feature & Operations Guide • Release 2.0 (SRE Edition)

Document Scope: Comprehensive Feature & Architecture Guide	Target Audience: DevOps, SREs, Engineering Leads & Founders
Core Engine: Django 5 + SQLite WAL (<120MB RAM, zero DB overhead)	Monetization: Bachs SaaS Gateway (₦2,000 Indie Pro & ₦5,000 Team Pro)
Open Source: 100% MIT License (Free Self-Hosting)	Live Platform URL: https://pingbud.us (Commercial Cloud)

1. Executive Overview & Core Engineering Principles

PulseWatch (commercially deployed at pingbud.us) is a high-class, ultra-reliable synthetic uptime monitoring and incident response platform. It was engineered to solve the prohibitive foreign exchange barrier and high operational complexity of legacy monitoring tools like Pingdom, Datadog Synthetics, and BetterStack. Where legacy SaaS tools cost $35 to $80/mo (₦55,000 to ₦125,000/mo), PulseWatch delivers enterprise-grade multi-region consensus probing, automated AI root-cause analysis (RCA), and multi-channel alerting for ₦2,000/mo ($1.30/mo) or 100% free self-hosting.

- Single-Process SQLite WAL Engine: Eliminates multi-node PostgreSQL, Redis, or managed timeseries database costs. Utilizes SQLite in Write-Ahead Logging mode (PRAGMA journal_mode=WAL;) with concurrent, lock-free readers. Consumes under 120 MB RAM, easily running on a $2.50/mo VPS or free container tier.
- Zero False Alarm SRE Guarantee: Outages are never declared from a single network hiccup. Multi-step retries with exponential jitter and cross-region consensus probes verify failures before triggering notification channels.
- Free Notification Infrastructure: Integrates native 100% free Telegram Bot alerting and Resend email delivery, keeping monthly communication overhead at zero.
- Native Nigerian Naira Billing (Bachs Gateway): Overcomes foreign card blocks with instant Providus Bank transfer virtual accounts (9928341029 Providus Bank) and debit card processing.

2. Platform Module & View Architecture Map

PulseWatch is organized into clean, dedicated modules accessible via a persistent liquid-glass sidebar:

Module / Route	Primary Purpose	Key Operational Features
Operational Dashboard /dashboard	Real-time infrastructure health command center.	Overall 24h uptime %, average latency ms, active incidents counter, down-first prioritized health probes list, 60-bucket historical bars, latency sparklines, TLS expiry countdowns, ⌘K command palette.
Scheduled Maintenance /maintenance	Planned downtime scheduling & SLA protection.	Create future maintenance windows, suppress false midnight alerts, protect SLA uptime calculations, and broadcast maintenance notices to public status pages.
Incidents & Post-Mortems /incidents	Outage timeline & root-cause analysis.	Automated AI Root-Cause Analysis (RCA), Mean Time to Resolution (MTTR) calculation, filter active vs. resolved incidents, 1-click Export CSV and Export JSON for client SLA audits.
Alert Channels & Routing channels view	Multi-channel notification routing.	Configure free Telegram Bots (@BotFather), Resend email alerts, Slack Incoming Webhooks, Discord embeds, generic webhooks (HMAC-SHA256 signed), and Termii/Twilio SMS.
Billing & Invoices bills view	Bachs subscription & payment receipts.	Transparent plans (Community ₦0, Indie Pro ₦2,000/mo, Team Pro ₦5,000/mo), Providus Bank instant virtual account transfer, 1-click printable PDF tax invoices, community donation fund.
Workspace Settings settings view	Team RBAC, Google OAuth & API keys.	Role-Based Access Control (Owner, Editor, Viewer), Link Google / Gmail account for 1-click login, Resend email configuration with live test dispatch, quiet hours scheduler, scoped API keys.
Public Status Page /status	Customer-facing service transparency.	Live headline status ('All Systems Operational'), 90-day availability history, active incident updates, custom branding, and custom domain mapping (status.yourcompany.com).

3. Health Monitoring Engine & The 7 Probe Protocols

PulseWatch features a robust, multi-protocol synthetic probing engine capable of validating infrastructure health across the application, transport, network, and security layers. Each monitor runs on a configurable interval (10s, 30s, 60s, or custom) and continuously records latency percentiles, uptime percentages, and status history.

Probe Protocol	Target Scope	Configuration Parameters	Outage Triggers & Failure Detection
1. HTTP / HTTPS	Web endpoints, REST APIs, GraphQL, microservices.	Methods: GET, POST, HEAD, PUT, PATCH, DELETE. Custom JSON headers (e.g. Authorization), custom request payload body, expected status code matcher (e.g. 200, 201, 2xx, 3xx).	HTTP 4xx/5xx status codes, request timeouts, SSL handshake aborts, truncated responses, DNS failures.
2. Keyword Matcher	Application layer semantic validation.	Modes: contains or not_contains. Regex or plain substring search across full HTTP response body.	Catches subtle application failures returning HTTP 200: database connection error strings, maintenance screens, white-screen-of-death, missing UI markers.
3. ICMP Ping	Network host reachability & router availability.	Raw socket ICMP echo packets, configurable timeout (1s to 10s), packet count, and loss threshold.	BGP routing failures, packet loss >50%, host crashes, severe network latency spikes.
4. TCP Socket	Database & service daemon connectivity.	Host and arbitrary port: PostgreSQL (5432), Redis (6379), MySQL (3306), SSH (22), SMTP (587), Kafka (9092).	Connection refused (ECONNREFUSED), port timeouts, process crashes, host firewall blocking.
5. TLS / SSL Validity	Certificate validity & expiration warnings.	SNI hostname validation. Automatic warning alerts at 30, 14, 7, and 1 day before certificate expiration.	Expired certificates, untrusted CA root authorities, hostname mismatches, revoked certificates, self-signed certificates.
6. DNS Resolution	Domain Name System record integrity.	Record types: A, AAAA, CNAME, MX, TXT. Expected resolved address matcher, nameserver selection.	DNS hijacking, domain registrar expiration, misconfigured nameservers, NXDOMAIN lookup errors.
7. Cron Heartbeat	Background cron jobs & dead-man switch.	Unique ingestion endpoint (e.g. curl -fsS /api/heartbeat/<token>/). Configurable grace period (1 to 60 mins).	Silent script crashes, OOM killer terminations, hung background worker queues, missed cron schedules.

Multi-Step Verification & Jittered Retry Execution Rules

To maintain strict SLA integrity, probe executions adhere to rigorous automated verification standards:

- Asynchronous Non-Blocking Engine: Synthetic probes execute via lightweight background threads with tight socket timeouts (default 5.0s), preventing hung targets from blocking monitoring loops.
- Jittered Retry Loop: When a probe fails, PulseWatch triggers an immediate secondary verification with randomized jitter (1s–3s). Only when consecutive retries fail is the incident pipeline activated.
- Historical 60-Bucket Rollups: Response latencies are aggregated into 60 chronological buckets displayed as sparklines and colored uptime pills (Emerald = Up, Rose = Down, Amber = Degraded).

SECURITY SURFACE: INTERNAL SUBNET SHIELD

SSRF Protection Architecture: PulseWatch actively prevents Server-Side Request Forgery. When adding or updating monitors, the probe dispatcher checks resolved IP addresses against private and cloud-internal subnets (127.0.0.0/8, 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16, and AWS metadata 169.254.169.254). Internal addresses are automatically blocked to prevent probing of host-level infrastructure.

4. Multi-Region Consensus Probing

Single-node monitoring tools suffer from a fatal flaw: local network blips, ISP routing issues, or CDN edge cache flushes cause false alarms that wake engineers at 3 AM. PulseWatch solves this through Multi-Region Consensus Probing:

- Global Edge Verification: On Indie Pro and Team Pro tiers, probes originate from multiple geographically distributed nodes (Lagos, London/Frankfurt EU, and US-East).
- Consensus Thresholds: An incident is only declared when at least 2 of 3 (or a majority of edge probes) independently confirm the endpoint is unreachable. If only 1 node fails while others succeed, the incident is tagged as a localized transit blip and suppressed from waking engineers.
- Localized CDN Drop Detection: Enables SREs to distinguish between an actual origin server crash versus an edge routing anomaly affecting only a single geographic region.

5. Alert Channels & Multi-Channel Notification Routing

PulseWatch decouples health checks from alert routing. Notifications can be configured globally or targeted per monitor:

Notification Channel	Carrier / Protocol	Setup & Parameters	Delivery Cost
Telegram Bot	Telegram Bot API (Direct HTTPS)	Enter Bot Token (created via @BotFather) and target Chat ID (group or channel). Markdown alert cards with instant 1-click incident status buttons.	100% Free Forever
Resend Email	Resend REST API (resend.com)	Configured with RESEND_API_KEY. Sends formatted HTML incident notices with AI diagnostic summaries and recovery confirmations. Includes live test dispatch in Settings.	Free Tier (3k/mo)
Slack	Incoming Webhook / Block Kit	Paste Slack Incoming Webhook URL. Rich colored attachments (Red for Down, Green for Recovered) with latency statistics and affected monitor link.	100% Free
Discord	Discord Server Webhook	Paste Discord Channel Webhook URL. Embedded rich alerts featuring timestamp, response code, and root-cause diagnostic preview.	100% Free
Generic Webhook	HTTP POST with HMAC-SHA256	Endpoint URL and optional Secret Signing Key. Payloads contain JSON structured event data for CI/CD pipelines, auto-remediation scripts, or PagerDuty automations.	100% Free
SMS / WhatsApp	Termii (Africa) & Twilio	API key & sender ID. Dispatches direct SMS to on-call phone numbers for emergency tier-1 production outages.	₦3.50/SMS (Pay-as-you-go)

Quiet Hours, Escalation Delays & Notification Hygiene

Alert fatigue causes missed outages. PulseWatch includes built-in operational hygiene controls in Workspace Settings:

- Quiet Hours Scheduler: Define a daily quiet period (e.g. 23:00 to 07:00 WAT). Non-critical alerts are buffered until quiet hours end.
- Escalation Policy & Tiered Paging: Configure an escalation delay (e.g. 5 minutes). When an outage begins, PulseWatch first notifies Slack or Telegram. If the incident remains unacknowledged after the threshold, it escalates to phone SMS or high-priority email.
- Automatic Recovery Broadcast: When the probe recovers and succeeds for two consecutive checks, an automatic 'Recovered' notice is dispatched across all channels with the total downtime duration.

6. Automated Incident Management & AI Root-Cause Analysis (RCA)

When an outage is verified, PulseWatch transitions immediately into incident response mode. SRE teams access a centralized incident management center (accessible at /incidents):

- Automated AI Root-Cause Analysis (RCA): An embedded diagnostic engine parses failed HTTP headers, response snippets, network error strings, and TLS handshakes to generate concise, human-readable explanations. For example: 'Upstream 502 Bad Gateway: NGINX reverse proxy cannot connect to Gunicorn socket on 127.0.0.1:8000. Process likely terminated due to memory exhaustion.'
- Mean Time to Resolution (MTTR) SRE Metrics: Automatically computes overall and per-monitor MTTR (in minutes), providing engineering management with clear reliability benchmarks across release cycles.
- Incident Search & Filtering: Filter between currently active outages vs. resolved historical incidents, and search incidents instantly by affected service, failure cause, or AI diagnostic summary.
- 1-Click Audit Export (CSV & JSON): Download full historical incident logs with timestamps, durations, causes, and post-mortem summaries via /api/incidents/export/?format=csv or json for customer SLA reporting and post-incident reviews.

7. Scheduled Maintenance Windows & SLA Protection

A critical flaw of basic monitoring services is recording false downtime during planned deployments or database maintenance. PulseWatch includes a dedicated Scheduled Maintenance subsystem (accessible at /maintenance):

Capability	How It Operates	SRE Benefit
Alert Suppression	During the window (defined by Start and End timestamps), all Telegram, Email, and Slack notifications are automatically silenced for associated monitors.	No false alarms waking on-call engineers during planned midnight database schema migrations or software deployments.
SLA Protection	Downtime occurring within an active maintenance window is tagged as maintenance_downtime and excluded from negative uptime % penalties.	Preserves contractual 99.9% uptime SLA commitments for enterprise and government clients.
Public Notice Broadcast	The public status page (/status) dynamically renders a prominent maintenance banner with schedule and estimated completion time.	Informs customers and stakeholders in advance, dramatically reducing inbound support tickets.
Granular Targeting	Windows can be applied workspace-wide or scoped to specific monitor subsets (e.g. only Payment API while Authentication API remains active).	Prevents masking unrelated outages in unmaintained microservices.

OPERATIONAL LIFECYCLE: AUTOMATED TIME GATING

Maintenance Window Lifecycle: Maintenance windows progress through three automated states: 1. Upcoming: Scheduled in advance, visible on the dashboard and status page. 2. Active: Probes continue checking for data, but notifications are suppressed and SLA calculations are shielded. 3. Past: Automatically completes when the end timestamp expires, with normal alerting immediately restored.

8. Authentication, Team RBAC & Workspace Administration

PulseWatch provides multi-user enterprise governance with Role-Based Access Control (RBAC), self-service password recovery, and Google OAuth integration:

Role	Scope	Permitted Capabilities	Enforced Restrictions
☐ Owner	Full Administrative Control	Manage Bachs billing subscriptions, generate/revoke scoped API keys, invite/remove team members, delete monitors, configure quiet hours, trigger SQLite WAL checkpoints.	None (super-administrator).
☐☐ Editor	Read & Write DevOps Access	Create, edit, pause, and resume health monitors; schedule maintenance windows; test alert channels; update status page announcements.	Cannot alter billing plans, invite owners, revoke admin API keys, or delete the workspace.
☐☐ Viewer	Read-Only Stakeholder Access	Inspect real-time metrics, probe sparklines, incident logs, public status pages, and export CSV/JSON incident reports.	All modification, probe execution, creation, and deletion requests return HTTP 403 Forbidden.

Identity & Security Subsystems

- Self-Service Password Recovery via Resend: Users who misplace credentials can request a password reset from the login screen or modal. A cryptographically signed token (backed by Django's default_token_generator) is dispatched via Resend, allowing secure, verified password updates.
- Google OAuth & Gmail Account Linking: Users can link their Google / Gmail account directly inside Workspace Settings (General tab) for seamless 1-click Google authentication.
- Developer API Keys: Generate scoped API tokens (read, write, admin) for Terraform, GitHub Actions, or automated probe provisioning with standard Authorization: Token <key> headers.

9. Billing, Bachs SaaS Gateway & Community Sponsorship

PulseWatch breaks the foreign exchange barrier with native Nigerian Naira (NGN) pricing powered by Bachs (bachs.io):

Plan Tier	Price (NGN)	Check Frequency	Included Quotas & Capabilities
Community Free	₦0 / forever	60s to 3 mins	Up to 10 monitors, local node probing, Telegram & Slack alerts, 30-day retention, 1 admin seat.
Indie Pro	₦2,000 / month	☐ 30 Seconds	Up to 50 monitors, 3 global consensus regions (Lagos, US, EU), unlimited AI RCA, 1-year metric retention, 5 team seats, custom status domain.
Team Pro	₦5,000 / month	☐ 10 Seconds	Unlimited monitors, 6 global edge regions, unlimited seats with full RBAC, priority SMS escalation, multi-year retention, white-labeled status pages.

Bachs Checkout, Printable Invoices & Community Sponsorship

- Providus Bank Transfer: Instant virtual account transfer (Account: 9928341029 Providus Bank). Automatically upgrades the workspace the moment the transfer is confirmed.
- Debit Card Processing: Seamless debit card checkout via Bachs with 1-click renewal.
- Printable Tax Invoices: Every billing event produces an official, printable PDF tax invoice (e.g. INV-2026-9403) for enterprise accounting and tax audits.
- Open-Source Community Sponsorship: A dedicated donation modal allows community members to sponsor edge probe infrastructure with presets (₦1k, ₦2k, ₦5k, ₦10k) or custom contributions.

10. Public Status Pages & Custom Domain Branding

PulseWatch includes a standalone, high-performance public status page (accessible at /status) designed to keep users informed during outages and maintain public trust:

- Live Headline Status: Displays 'All Systems Operational' (Emerald) or 'Active Outage Detected' (Rose) with real-time updates.
- Interactive 90-Day Uptime Bars: Daily availability percentage bars with tooltip latency metrics.
- Active Incident Feed & Maintenance Notices: Real-time incident logs with investigator notes and scheduled maintenance broadcasts.
- Custom Domain Mapping: Map status pages to your own domain (e.g. status.yourcompany.com) via a simple CNAME record.

11. 100% Free Hosting & Zero-Cost Deployment Blueprint

Because of PulseWatch's single-node SQLite WAL architecture (<120MB RAM), it can be hosted 100% free forever:

Hosting Platform	Free Allowance	Deployment Configuration	Zero-Downtime Architecture
Render.com	Free Web Service (512MB RAM)	Connect GitHub repo, Python 3.11, command: gunicorn pulsewatch.wsgi.	Since PulseWatch continuously executes synthetic checks, its own outgoing pings prevent Render from spinning down!
Fly.io	Up to 3 free shared-cpu VMs	Deploy with fly launch using the included Dockerfile. Attach a 1GB persistent volume for SQLite.	Ultra-fast worldwide edge deployment with zero database maintenance.
Koyeb	1 Free Nano Service (512MB RAM)	Direct git-push deployment with automatic HTTPS domain.	Continuous background execution with zero idle sleep mode.
Neon / Supabase	Free Serverless Postgres (0.5GB)	Optional: set DATABASE_URL to connect external managed PostgreSQL.	Only required if you prefer managed Postgres over native SQLite WAL.

12. SRE Operational Quick Reference Endpoints

Resource / Action	Method & Endpoint	Purpose & Expected Response
Public Status Page	GET /status or /api/status-page/	Public availability dashboard, active incidents, and 90-day uptime bars.
Cron Heartbeat Ingest	GET /api/heartbeat/<token>/	Dead-man switch ingest endpoint for background workers and cron jobs.
Prometheus Scrape	GET /metrics/	Prometheus-compatible scrape endpoint for Grafana dashboards.
Incident Audit Export	GET /api/incidents/export/?format=csv|json	Download full historical outage records, AI root-cause summaries, and MTTR data.
On-Demand Probe Check	POST /api/monitors/<id>/check/	Triggers immediate probe execution without waiting for the next scheduled interval.
Password Reset Dispatch	POST /api/auth/forgot-password/	Dispatches cryptographic password reset token via Resend.
Bachs Checkout Session	POST /api/bachs/checkout/	Generates Providus Bank transfer account and card checkout session.

PULSEWATCH RELEASE 2.0 • PRODUCTION READY

System Summary & Production Sign-Off: PulseWatch delivers the highest reliability-to-cost ratio in synthetic monitoring. With single-node SQLite WAL performance (<120MB RAM), free Telegram Bot and Resend email alerts, automated AI root-cause analysis, and Bachs Nigerian Naira pricing (₦2,000/mo), it provides world-class SRE infrastructure accessible to everyone. Source code and updates available at https://pingbud.us.